

NKPC HSA Bayesian State-Space Estimation

June 22, 2026

1 Data and Transformations

Raw data are read from `data/`. Processed model-ready data are written to `data/processed/`. The HSA competition aggregator convention is $N_t = 100 \log(\text{level}_t)$ unless a run metadata file explicitly states that the series was already transformed.

2 Model and Unit Conventions

Kappa-related priors are specified in physical units. Some samplers use internally scaled parameters with `KAPPA_SCALE=100`; all posterior draws, tables, figures, Chib ordinates, and SDDR calculations are reported in physical units.

3 Posterior Summaries

note

Run scripts/06_make_tables_figures.py first.

4 MCMC Diagnostics

note

Run scripts/03_run_diagnostics.py first.

5 Prior Sensitivity

note

Run scripts/04_prior_sensitivity.py first.

6 Model Comparison

note

Run scripts/05_model_comparison.py first.

7 Figures

Posterior predictive checks are generated after model runs.